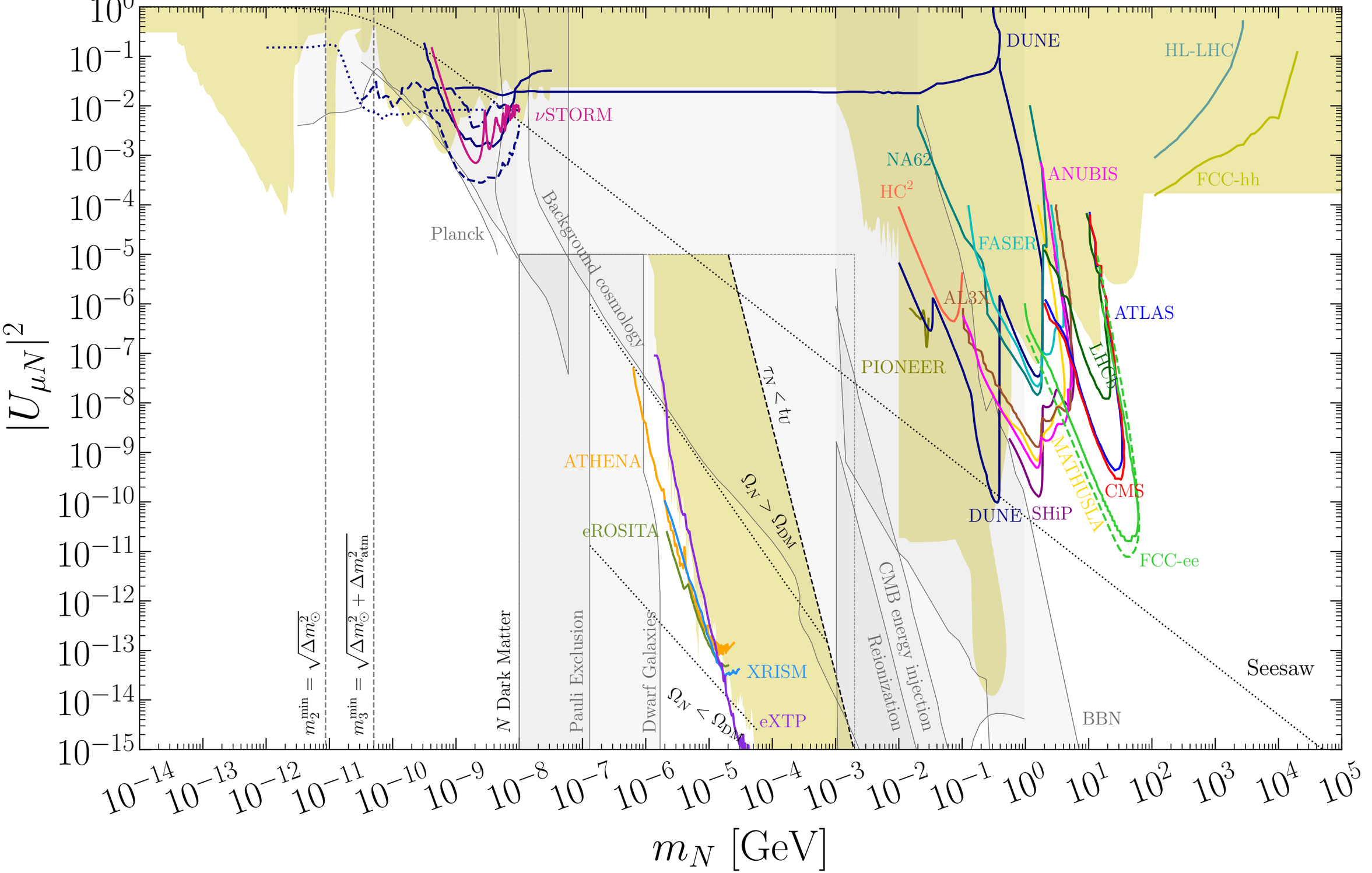

$|U_{\mu N}|^2$
m_N [GeV]
DUNE
HL-LHC
νSTORM
NA62
HC^2
ANUBIS
FCC-hh
FASER
Planck
Background cosmology
AL3X
ATLAS
PIONEER
LHCb
$\tau_N < t_U$
ATHENA
MATHUSLA
CMS
$\Omega_N > \Omega_{\rm DM}$
DUNE
SHiP
eROSITA
FCC-ee
CMB energy injection
Reionization
XRISM
Seesaw
$\Omega_N < \Omega_{\rm DM}$
eXTP
BBN
$m_2^{\rm min} = \sqrt{\Delta m^2_\odot}$
$m_3^{\rm min} = \sqrt{\Delta m^2_\odot + \Delta m^2_{\rm atm}}$
N Dark Matter
Pauli Exclusion
Dwarf Galaxies